

Bupivacaine: Drug information - UpToDate

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For additional information see "Bupivacaine: Patient drug information" and "Bupivacaine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Obstetrical anesthesia (injection only):

There have been reports of cardiac arrest with difficult resuscitation or death during use of bupivacaine for epidural anesthesia in obstetrical patients. In most cases, this has followed use of the 0.75% (7.5 mg/mL) concentration. Resuscitation has been difficult or impossible despite apparently adequate preparation and appropriate management. Cardiac arrest has occurred after convulsions resulting from systemic toxicity, presumably following unintentional intravascular injection. The 0.75% (7.5 mg/mL) concentration of bupivacaine is not recommended for obstetrical anesthesia and should be reserved for surgical procedures where a high degree of muscle relaxation and prolonged effect are necessary.

Risk of potential adverse embolic effects resulting from inadvertent intravascular injection (subacromial injection only):

Inadvertent intravascular injection could cause bupivacaine droplets to be deposited in the pulmonary and other capillary beds. Administer bupivacaine into the subacromial space at the end of arthroscopic shoulder surgery. Direct arthroscopic visualization must be used to confirm proper placement of the needle tip before injecting bupivacaine.

Brand Names: US

- BUPivacaine Fisiopharma;
- BUPivacaine Spinal;
- Marcaine;
- Marcaine Preservative Free;
- Marcaine Spinal;
- Posimir [DSC];
- Sensorcaine;
- Sensorcaine-MPF;
- Xaracoll [DSC]

Brand Names: Canada

- Marcaine;

- Marcaine Spinal [DSC];
- Sensorcaine

Pharmacologic Category

- Local Anesthetic

Dosing: Adult

Analgesia or anesthesia

Analgesia or anesthesia:

Implant:

Analgesia, postsurgical: 300 mg (3 × 100 mg implants) as a single dose placed into the inguinal hernia repair surgical site.

Injection:

Note: Dose varies with procedure, depth of anesthesia, vascularity of tissues, duration of anesthesia, and condition of patient. Do not use solutions containing preservatives for caudal or epidural block.

Local anesthesia: Infiltration: 0.25% infiltrated locally; maximum: 175 mg. **Note:** Aspiration should be performed prior to each injection; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref).

Caudal block (preservative free): 15 to 30 mL of 0.25% or 0.5%

Epidural block (other than caudal block; preservative free): Administer in 3 to 5 mL increments, allowing sufficient time to detect toxic manifestations of inadvertent IV or intrathecal administration: 10 to 20 mL of 0.25% or 0.5%

Surgical procedures requiring a high degree of muscle relaxation and prolonged effects **only: Note:** Not to be used in obstetrical cases: 10 to 20 mL of 0.75%

Peripheral nerve block: 5 mL of 0.25% or 0.5%; maximum: 400 mg/day

Sympathetic nerve block: 20 to 50 mL of 0.25%

Retrobulbar anesthesia: 2 to 4 mL of 0.75%

Spinal anesthesia: Preservative free solution of 0.75% bupivacaine in 8.25% dextrose:

Lower extremity and perineal procedures: 1 mL

Lower abdominal procedures: 1.6 mL

Labor and normal vaginal delivery: 0.8 mL (higher doses may be required in some patients)

Cesarean delivery: 1 to 1.4 mL

Combined spinal-epidural (CSE) technique for labor analgesia (off-label dosing [spinal component]): 1.75 to 2.5 mg combined with fentanyl 15 mcg (Ref).

Combined spinal-epidural (CSE) technique for anesthesia for Cesarean delivery (off-label dosing [spinal component]): 8 to 12 mg combined with fentanyl 15 mcg; in addition to fentanyl, may also include a longer-acting opioid (ie, morphine 100 to 150 mcg) for postoperative analgesia (Ref).

Subacromial injection (Posimir): Infiltration: Arthroscopic subacromial decompression surgery: 660 mg (5 mL) as a single dose.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no specific dosage adjustments provided in the manufacturer's labeling; however, need for dosage adjustment is unlikely as bupivacaine is primarily hepatically metabolized to inactive metabolites (Ref).

Dosing: Liver Impairment: Adult

Implant: There are no specific dosage adjustments provided in the manufacturer's labeling. Risk of toxicity may be increased in patients with severe impairment; consider more frequent monitoring for toxicities in patients with moderate or severe impairment.

Injection: There are no specific dosage adjustments provided in the manufacturer's labeling. Risk of toxicity may be increased in patients with severe impairment; consider a reduced dose and more frequent monitoring for toxicities in patients with moderate or severe impairment.

Subacromial injection (Posimir): There are no specific dosage adjustments provided in the manufacturer's labeling. Risk of toxicity may be increased in patients with severe impairment; consider a reduced dose and more frequent monitoring for toxicities in patients with moderate or severe impairment.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Bupivacaine: Pediatric drug information")

Note: Dose varies with procedure, depth of anesthesia, vascularity of tissues, duration of anesthesia, and condition of patient. Preservative-free formulations are recommended for administration into the CNS space (eg, epidural, caudal, spinal). Consider incremental administration with negative aspiration prior to each injection; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref). Should only be administered under the supervision of a qualified physician experienced in the use of anesthetics. In obese pediatric patients, the preferred weight used for dose calculation is undefined and specific pediatric data are sparse (Ref); some have suggested the use of lean body mass/weight not ideal body weight (IBW) (Ref).

Central nerve block/anesthesia

Central nerve block/anesthesia:

Caudal block: Reported dosing variable based on procedure; dependent on necessary dermatome level and corresponding volume of space:

Infants and Children: Limited data available: Usual concentration $\leq 0.25\%$ solution with or without epinephrine: Usual reported dose range: 0.5 to 1.3 mL/kg (maximal volume of drug: 20 mL); dose should not exceed 2 **mg/kg** plain solution, or 3 **mg/kg** with epinephrine. In infants, routine use of concentrations $\leq 0.25\%$ have been suggested to reduce risk of bupivacaine cardiotoxicity (Ref).

Adolescents: Usual concentration 0.25% to 0.5% solution with or without epinephrine: 15 to 30 mL (Ref).

Epidural block: Reported dosing variable; among other factors, dose dependent on necessary dermatome level and corresponding volume of epidural space (Ref):

Infants: Limited data available: Usual concentration $\leq 0.25\%$ with or without epinephrine: 0.7 to 0.75 mL/kg; maximum dose: 2.5 **mg/kg**; **Note:** For infants (particularly young infants), if repeat injections necessary, a decreased dose may be necessary to prevent drug accumulation. Some experts suggest if at least 45 minutes since initial dose, reduce dose to $\frac{1}{3}$ of the initial or if at least 90 minutes since initial dose, then reduce dose to half of the initial. If additional doses are necessary, doses should be reduced to half of the previous dose (Ref).

Children: Limited data available: Usual concentration 0.25% solution: Initial: 0.3 to 0.6 mL/kg (maximal volume of drug: 20 mL); maximum dose: 2.5 **mg/kg** (Ref).

Adolescents: 0.25% or 0.5% solution: 10 to 20 mL administered in 3 to 5 mL increments; if high degree of muscle relaxation and prolonged effects needed, may consider 0.75% solution: 10 to 20 mL (Ref).

Epidural, continuous infusion: Limited data available in infants and children <12 years (Ref): **Note:** Use has generally been replaced by other agents (eg, ropivacaine) (Ref).

Loading dose: Usual concentration: 0.25% solution: 2 to 2.5 **mg/kg**.

Infusion:

Infants <6 months: 0.2 **mg/kg/hour**.

Infants ≥6 months: 0.25 **mg/kg/hour**.

Children and Adolescents: 0.3 **mg/kg/hour**; range: 0.2 to 0.4 **mg/kg/hour**.

Local anesthesia

Local anesthesia: Infants, Children, and Adolescents: Limited data in infants and children: Usual concentration 0.25% solution: Infiltrate local area; maximum dose in infants and children: 2.5 **mg/kg** or 150 mg, whichever is less; maximum dose in adolescents: 175 mg (Ref).

Peripheral nerve block

Peripheral nerve block: Limited data available in infants and children: **Note:** Dose varies with location of block (ie, procedure), depth of anesthesia, vascularity of tissues, duration of anesthesia, and condition of patient.

Infants ≥6 months and Children: Usual concentration 0.125% or 0.25% solution with or without epinephrine: The volume of dose (mL/kg) and concentration of solution are site specific based upon anatomy and variable among patients and procedure; see below ranges. For infants <6 months, maximum doses should be reduced by 30% (Ref). Maximum dose plain solution: 2 **mg/kg** or 150 mg, whichever is less, or maximum dose with epinephrine: 3 **mg/kg** or 200 mg of bupivacaine, whichever is less.

Commonly suggested doses (Ref):

Head and neck: 0.05 mL/kg.

Upper extremity:

Brachial plexus: 0.2 to 0.3 mL/kg.

Digital nerve: 0.05 mL/kg.

Truncal blocks:

Transversus abdominis plane: 0.2 to 0.5 mL/kg (Ref).

Rectus sheath: 0.1 mL/kg.

Ilioinguinal: 0.075 mL/kg.

Lower extremity blocks:

Femoral nerve: 0.2 to 0.3 mL/kg.

Sciatic nerve: 0.2 to 0.3 mL/kg.

Adolescents: 0.25% or 0.5% solution with or without epinephrine: 5 mL; maximum daily dose: 400 **mg/day** (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Implant:

>10%: Local: Edema at insertion site (15%)

1% to 10%:

Gastrointestinal: Dysgeusia (8%)

Genitourinary: Scrotal edema (3%)

Nervous system: Headache (4%)

Neuromuscular & skeletal: Tremor (4%)

Ophthalmic: Blurred vision (4%)

Miscellaneous: Fever (2%), seroma (3%)

Frequency not defined: Infection: Wound infection

Injection:

Reactions listed are based on reports for bupivacaine, bupivacaine/epinephrine, and/or other local anesthetics.

>10%:

Dermatologic: Pruritus (3% to 19%; including eye pruritus and anorectal pruritus)

Gastrointestinal: Constipation (7% to 42%), dysgeusia (15% to 18%), nausea (39% to 60%, including post-procedural nausea), vomiting (10% to 29%)

Nervous system: Dizziness (21% to 40%), drowsiness (32% to 41%), headache (6% to 32%), paresthesia (4% to 18%)

Otic: Tinnitus (13%)

1% to 10%:

Cardiovascular: Angina pectoris (<2%), flattened T wave on ECG (<2%), hypertension (4%), inversion T wave on ECG (4%), peripheral edema (4%)

Dermatologic: Injection site pruritus (3%), skin rash (4%)

Gastrointestinal: Upper abdominal pain (1%)

Genitourinary: Dysmenorrhea (3%), urinary retention (1% to 3%)

Hematologic & oncologic: Bruise (3%)

Nervous system: Fatigue (<2%), hypoesthesia ($\leq$ 17%), insomnia (3% to 7%)

Neuromuscular & skeletal: Limb pain (3%), muscle twitching (4%), musculoskeletal pain (5%)

Ophthalmic: Blepharospasm (<2%)

Respiratory: Cough (2%), dyspnea (4%), exacerbation of pulmonary arterial hypertension (<2%), nasal congestion (3%), oropharyngeal pain (1%)

Miscellaneous: Fever (6% to 9%)

Frequency not defined:

Cardiovascular: Bradycardia, cardiac insufficiency, circulatory shock, heart block, hypotension, low cardiac output, ventricular arrhythmia

Gastrointestinal: Fecal incontinence, loss of anal sphincter control

Genitourinary: Prolonged labor, sexual disorder (loss of function), urinary incontinence

Hematologic & oncologic: Methemoglobinemia

Hypersensitivity: Hypersensitivity reaction

Infection: Septic meningitis

Nervous system: Anxiety, arachnoiditis, central nervous system depression, central nervous system stimulation, chills, coma, confusion, cranial nerve palsy, localized numbness (perineal), loss of consciousness, meningism, paralysis, paraplegia, persistent anesthesia, restlessness, seizure, shivering

Neuromuscular & skeletal: Asthenia, back pain, lower extremity weakness, tremor

Ophthalmic: Blurred vision, miosis

Respiratory: Apnea, hypoventilation, respiratory depression, respiratory paralysis

Contraindications

Hypersensitivity to bupivacaine hydrochloride, amide-type local anesthetics, or any component of the formulation; obstetrical paracervical block anesthesia; IV regional anesthesia (Bier block) (injection only).

Canadian labeling: Additional contraindications (not in US labeling): Severe shock and in heart block where there is inflammation and/or sepsis near the proposed injection site (Marcaine only).

Warnings/Precautions

Concerns related to adverse effects:

- Cardiovascular effects: Bupivacaine-containing products have been associated with rare occurrences of arrhythmias, cardiac arrest, and death.
- Intra-articular infusion related chondrolysis: Continuous intra-articular infusion of local anesthetics after arthroscopic or other surgical procedures is **not** an approved use; chondrolysis (primarily shoulder joint) has occurred following infusion, with some patients requiring arthroplasty or shoulder replacement.
- Methemoglobinemia: Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months of age are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, light-headedness, fatigue).
- Respiratory arrest: Local anesthetics have been associated with rare occurrences of sudden respiratory arrest, especially when administered near the head or neck.
- Seizures: Convulsions due to systemic toxicity leading to cardiac arrest have also been reported, presumably following unintentional intravascular injection or administration near the head or neck.

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular disease including patients with hypotension or heart block.
- Hepatic impairment: Use with caution in patients with hepatic impairment.

Special populations:

- Acutely ill patients: Use with caution in acutely ill patients; dose reduction may be required.
- Debilitated patients: Use with caution in debilitated patients; dose reduction may be required.
- Older adult: Use with caution in the elderly; dose reduction may be required.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP 1997; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.
- Interchangeability: Subacromial injection is not interchangeable with other bupivacaine formulations; bioequivalence is not the same even when milligram dosage is the same.
- Obstetrical anesthesia: The bupivacaine 0.75% concentration is not recommended for obstetrical anesthesia. There have been reports of cardiac arrest with difficult resuscitation or death during use of bupivacaine for epidural anesthesia in obstetrical patients.
- Preservative-containing solutions: **Do not use solutions containing preservatives for caudal or epidural block.**
- Sodium metabisulfite: May contain sodium metabisulfite; use caution in patients with asthma or a sulfite allergy.

Other warnings/precautions:

- Administration:
 - Injection: Use the lowest effective dose and fractional (incremental) doses when possible; repeat administration may result in accumulation of bupivacaine and metabolites. Intravascular and intrathecal injections should be avoided; aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular or intrathecal injection has been avoided. IV regional anesthesia (Bier block) is contraindicated; cardiac arrest and death have occurred with this method of administration.
 - Subacromial injection: For infiltration into the subacromial space only; not for epidural, intra-articular, intrathecal, intravascular, or regional nerve block administration. Avoid inadvertent intravascular administration; cardiac arrest and seizures have occurred. May be used with implantable materials (eg, polypropylene, polyester) and gut, nylon, polydioxanone, polypropylene, polyglycolic acid, and silk sutures.
- Test dose: A test dose is recommended prior to epidural administration (prior to initial dose) and all reinforcing doses with continuous catheter technique.
- Trained personnel: Clinicians using local anesthetic agents should be well trained in diagnosis and management of emergencies that may arise from the use of these agents. Resuscitative equipment, oxygen, and other resuscitative drugs should be available for immediate use.

Warnings: Additional Pediatric Considerations

Infants may be at greater risk for bupivacaine toxicity because α_1 -acid-glycoprotein concentration, the major serum protein to which bupivacaine is bound, is lower in neonates and infants compared with older children leading to an increase of free fraction of local anesthetic; use epidural infusions with caution in neonates and infants and monitor closely. Fat emulsion has been used to manage local anesthetic toxicity (refer to Lipid Emulsion (Plant Based) monograph for additional information) (McCloskey 1992).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Implant, Implant:

Xaracoll: 300 mg (3 x 100 mg) (3 ea [DSC])

Solution, Injection:

Posimir: 660 mg/5 mL (5 mL [DSC]) [contains benzyl alcohol]

Solution, Injection, as hydrochloride:

Marcaine: 0.25% (50 mL); 0.5% (50 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Sensorcaine: 0.25% (50 mL); 0.5% (50 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Generic: 0.25% (50 mL); 0.5% (30 mL, 50 mL)

Solution, Injection, as hydrochloride [preservative free]:

BUPIvacaine Fisiopharma: 2.5 mg/mL (5 mL, 10 mL); 0.5% (5 mL, 10 mL)

Marcaine: 0.75% (10 mL, 30 mL [DSC])

Marcaine Preservative Free: 0.25% (10 mL, 30 mL); 0.5% (10 mL, 30 mL); 0.75% (30 mL)

Sensorcaine-MPF: 0.25% (10 mL, 30 mL); 0.5% (10 mL, 30 mL); 0.75% (10 mL, 30 mL) [methylparaben free]

Generic: 0.25% (10 mL, 30 mL); 0.5% (10 mL, 30 mL); 0.75% (10 mL, 30 mL)

Solution, Intrathecal, as hydrochloride [preservative free]:

BUPIvacaine Spinal: 0.75% [7.5 mg/mL] (2 mL)

Marcaine Spinal: 0.75% [7.5 mg/mL] (2 mL)

Generic: 0.75% [7.5 mg/mL] (2 mL)

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (BUPIvacaine HCl (PF) Injection)

0.25% (per mL): \$0.25 - \$15.80

0.5% (per mL): \$0.30 - \$0.63

0.75% (per mL): \$0.30 - \$0.66

Solution (BUPIvacaine HCl Injection)

0.25% (per mL): \$0.09 - \$0.28

0.5% (per mL): \$0.07 - \$0.31

Solution (BUPIvacaine in Dextrose Intrathecal)

0.75-8.25% (per mL): \$4.71

Solution (Marcaine Injection)

0.25% (per mL): \$0.19

0.5% (per mL): \$0.15

0.75% (per mL): \$0.63

Solution (Marcaine Preservative Free Injection)

0.25% (per mL): \$0.53

0.5% (per mL): \$0.62

0.75% (per mL): \$0.33

Solution (Marcaine Spinal Intrathecal)

0.75-8.25% (per mL): \$0.87

Solution (Sensorcaine Injection)

0.25% (per mL): \$0.30

0.5% (per mL): \$0.34

Solution (Sensorcaine-MPF Injection)

0.25% (per mL): \$0.28

0.5% (per mL): \$0.30

0.75% (per mL): \$0.33

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Marcaine: 0.75% (20 mL)

Generic: 0.75% (10 mL, 20 mL, 30 mL)

Solution, Injection, as hydrochloride:

Marcaine: 0.5% (50 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Sensorcaine: 0.25% (10 mL, 20 mL); 0.5% (10 mL, 20 mL)

Generic: 0.25% (10 mL, 20 mL); 0.5% (10 mL, 20 mL); 0.75% (10 mL, 20 mL, 30 mL)

Solution, Intrathecal, as hydrochloride:

Marcaine Spinal: 0.75% [7.5 mg/mL] ([DSC])

Generic: 0.75% [7.5 mg/mL] (2 mL)

Administration: Adult

Implant: Each single-dose pouch is supplied as a sterile product that should be handled using aseptic technique. Oxygen and resuscitative medications, staff, and equipment should be immediately available. Implant is a ready-to-use product and only requires cutting the individual implants as needed to accommodate the surgical space. Aseptically peel outer pouch, remove and peel open inner blister pack; do not use scissors or scalpel to open the blister packaging. Inspect each implant prior to use; do not use if packaging has been compromised, or if implant appears discolored, contains foreign particles, or is collapsed, compressed, or misshapen; avoid excessive handling and compression. Aseptically cut each implant in half prior to placement into the surgical site; place 3 halves below the site of mesh placement and 3 halves just below the skin closure. Place dry implant into surgical site; when a topical antiseptic (eg, povidone iodine) is applied, allow surgical site to dry before implantation. Avoid contact with liquids prior to placement; pre-moistening may result in premature release of bupivacaine. Avoid additional local anesthetic administration within 96 hours

of inserting implant; toxic effects of local anesthetics are additive. If additional local anesthetic administration cannot be avoided, monitor patients for neurologic and cardiovascular effects related to local anesthetic systemic toxicity. Depending on clinical situation, consider surgical removal of implant if systemic toxicity occurs.

Injection: Solutions containing preservatives should not be used for epidural or caudal blocks. Addition of sodium bicarbonate to increase speed of onset may cause precipitation as pH approaches physiologic range (Ref). The On-Q infusion pump is used to slowly administer local anesthetics (eg, bupivacaine, lidocaine, ropivacaine) to or around surgical wound sites and/or in close proximity to peripheral nerves for postoperative analgesia. When infused directly into the shoulder, destruction of articular cartilage (chondrolysis) has occurred. On-Q pumps should never be placed directly into any joint.

Subacromial injection (Posimir): For infiltration into the subacromial space only following arthroscopic subacromial decompression surgery; not for epidural, intra-articular, intrathecal, intravascular, or regional nerve block administration. Do not use for pre-incisional or pre-procedural locoregional anesthetic techniques that require deep or complete sensory block. Do not dilute or mix with other drugs, diluents, or local anesthetics. Do not interchange with other bupivacaine formulations. Avoid use with other local anesthetics within 168 hours following administration. Draw up 5 mL into a syringe containing a large bore needle (≥ 16 gauge); discard needle. At the close of arthroscopic subacromial decompression surgery, administer entire dose into the subacromial space using an ≥ 18 -gauge needle. Insert needle through an existing arthroscopic port or intact skin to reach the subacromial space; confirm placement with direct arthroscopic visualization; avoid administering into the glenohumeral intra-articular space.

Administration: Pediatric

Parenteral: Solutions containing preservatives should not be used for epidural or caudal blocks; for epidural infusion, may use undiluted or diluted with preservative-free NS. Consider incremental administration with negative aspiration prior to each injection; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref).

Preparation for Administration: Pediatric

Epidural continuous infusion: Use preservative-free formulation, further dilute with preservative-free NS.

Storage/Stability

Implant: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Brief exposure to temperatures up to 40°C (104°F) may be tolerated provided the mean kinetic temperature does not exceed 25°C (77°F); however, such exposure should be minimized.

Injection: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F).

Subacromial injection: Store in original container at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Protect from light.

Use: Labeled Indications

Implant (Xaracoll): Placement into the surgical site to produce postsurgical analgesia for up to 24 hours following open inguinal hernia repair in adults.

Injection: Local or regional anesthesia; spinal anesthesia (0.75% in dextrose 8.25% injection); diagnostic and therapeutic procedures; obstetrical procedures (only 0.25% and 0.5% concentrations).

0.25%: Local infiltration, peripheral nerve block, sympathetic block, caudal or epidural block.

0.5%: Peripheral nerve block, caudal and epidural block.

0.75% (**not for obstetrical anesthesia**): Retrobulbar block, epidural block. **Note:** Reserve for surgical procedures where a high degree of muscle relaxation and prolonged effect are necessary.

Subacromial injection (Posimir): Infiltration administration in adults into the subacromial space under direct arthroscopic visualization to produce post-surgical analgesia for up to 72 hours following arthroscopic subacromial decompression.

Limitations of use: Safety and efficacy have not been established in other surgical procedures, including soft tissue surgical procedures, other orthopedic procedures, including for intra-articular administration, and boney procedures, or when used for neuraxial or peripheral nerve blockade.

Medication Safety Issues

Sound-alike/look-alike issues:

BUPIvacaine may be confused with mepivacaine, ROPivacaine

Marcaine may be confused with Narcan

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (conventional forms with liposomal counterparts; epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2C19 (Minor), CYP2D6 (Minor), CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Articaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

BUPIvacaine (Liposomal): BUPIvacaine may increase adverse/toxic effects of BUPIvacaine (Liposomal). Management: Bupivacaine may be administered immediately before, or administered in the same admixture syringe as liposomal bupivacaine as long as the ratio of the milligram dose of bupivacaine to liposomal bupivacaine does not exceed 1:2. *Risk D: Consider Therapy Modification*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

CycloPHOSphamide: May increase adverse/toxic effects of BUPIvacaine. Specifically, the risk of methemoglobinemia may be increased. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hyaluronidase: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Hydroxyurea: May increase adverse/toxic effects of BUPIvacaine. Specifically, the risk of methemoglobinemia may be increased. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Ifosfamide: May increase adverse/toxic effects of BUPIvacaine. Specifically, the risk of methemoglobinemia may be increased. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Local Anesthetics: May increase adverse/toxic effects of BUPIvacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant (Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nitrous Oxide: May increase adverse/toxic effects of BUPivacaine. Specifically, the risk of methemoglobinemia may be increased. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Propranolol: May increase serum concentrations of BUPivacaine. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

ROPivacaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Valproic Acid and Derivatives: May increase adverse/toxic effects of BUPivacaine. Specifically, the risk of methemoglobinemia may be increased. *Risk C: Monitor*

Pregnancy Considerations

Bupivacaine crosses the placenta.

Small amounts of bupivacaine can be found in maternal plasma following epidural, caudal, or pudendal nerve block anesthesia, and potentially causing varying degrees of maternal, fetal, and neonatal toxicity involving the CNS, peripheral vascular tone, and cardiac function.

Bupivacaine is approved for use at term in obstetrical anesthesia or analgesia. It is used as a component of regional anesthesia for labor analgesia and operative anesthesia, for intra-incisional analgesia at the end of a cesarean section, and for pudendal nerve blocks used in the second stage of labor or for repairing perineal lacerations (ACOG 2019). Use in obstetrical paracervical block anesthesia is contraindicated (may cause fetal bradycardia and death).

The bupivacaine 0.75% concentration is not recommended for obstetrical anesthesia. **There have been reports of cardiac arrest with difficult resuscitation or death during use of bupivacaine for epidural anesthesia in obstetrical patients. In most cases, this has followed use of the 0.75% concentration. Resuscitation has been difficult or impossible despite apparently adequate preparation and appropriate management. Cardiac arrest has occurred after convulsions resulting from systemic toxicity, presumably following unintentional intravascular injection. The 0.75% concentration should be reserved for surgical procedures where a high degree of muscle relaxation and prolonged effect are necessary.**

Breastfeeding Considerations

Bupivacaine and its active metabolite are present in breast milk.

Breast milk concentrations of bupivacaine were evaluated in a study of 27 women (~38 weeks gestation) administered bupivacaine 0.5% combined with lidocaine 2% for local anesthesia prior to cesarean delivery. Maternal serum and breast milk were sampled prior to, then at 2, 6, and 12 hours after the epidural infusion. Peak breast milk concentrations occurred 2 hours after the maternal dose for bupivacaine and 12 hours after the dose for the active metabolite. Adverse events were not observed in the breastfed infants (Ortega 1999). In a second study, bupivacaine was not detected in the serum of a breastfed 10-month old infant, feeding 4 times daily following maternal use of a continuous intrapleural bupivacaine 0.25% infusion (10 mL/hour) after a cholecystectomy. Breastfeeding resumed ~22 hours after the maternal infusion began; infant blood was sampled ~5 hours after a morning feed and ~52 hours after a maternal bolus dose (Baker 1989).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. However, the Academy of Breast Feeding Medicine considers local anesthetics such as bupivacaine compatible with breastfeeding. Elective surgery should be postponed until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or

hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Vital signs, state of consciousness; signs of CNS toxicity

Mechanism of Action

Blocks both the initiation and conduction of nerve impulses by decreasing the neuronal membrane's permeability to sodium ions, which results in inhibition of depolarization with resultant blockade of conduction

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Anesthesia (route and dose dependent):

Epidural: Up to 17 minutes to spread to T6 dermatome (Scott 1980).

Infiltration: Fast (Barash 2009); Dental injection: 2 to 10 minutes.

Spinal: Within 1 minute; maximum dermatome level achieved within 15 minutes in most cases.

Duration (route and dose dependent):

Epidural: 2 to 7.7 hours (Barash 2009).

Implant: Variable: Up to ~72 hours (Velanovich 2020).

Infiltration: 2 to 8 hours (Barash 2009); Dental injection: Up to 7 hours.

Spinal: 1.5 to 2.5 hours (Tsai 2007).

Subacromial injection: ~72 hours.

Distribution: V_d : Infants: 3.9 ± 2 L/kg; Children: 2.7 ± 0.2 L/kg.

Protein binding: 84% to 95%.

Metabolism: Hepatic; forms metabolite (pipecoloxylidine [PPX]).

Half-life elimination (age dependent): Implant: 19 hours; Injection: Neonates: 8.1 hours; Adults: 2.7 hours; Subacromial injection: 16.4 to 26.1 hours.

Time to peak, plasma: Caudal, epidural, or peripheral nerve block: 30 to 45 minutes; Implant: Median: 3 hours (range: 1.5 to 24 hours); Subacromial injection: Median: ~6 to 8 hours (range: 1 to ~27 hours).

Excretion: Urine (~6% unchanged).

Clearance: Infants: 7.1 ± 3.2 mL/kg/minute; Children: 10 ± 0.7 mL/kg/minute.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Elderly patients reached the maximal spread of analgesia and maximal motor blockade more rapidly than younger patients. Elderly patients also exhibited higher peak plasma concentrations following administration of this product. The total plasma clearance was decreased in these patients.

Patient Counseling Points

(For additional information see "Bupivacaine: Patient drug information")

What is this drug used for?

- It is used to numb an area before a procedure.
- It is used to ease pain after surgery.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Implant:

- Change in taste
- Headache

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

For all uses of this drug:

- Acidosis like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak
- Methemoglobinemia like blue or gray color of the lips, nails, or skin; a heartbeat that does not feel normal; seizures; very bad dizziness or passing out; very bad headache; feeling very sleepy; feeling tired or weak; or shortness of breath
- Fast, slow, or abnormal heartbeat
- Chest pain or pressure
- Dizziness or passing out
- Feeling lightheaded, sleepy, confused, or having blurred eyesight
- Change in balance
- Anxiety
- Change in speech
- Restlessness
- Shakiness
- Twitching
- Ringing in the ears
- Feeling nervous and excitable
- Trouble breathing, slow breathing, or shallow breathing
- Very upset stomach or throwing up
- Feeling hot or cold
- Sneezing
- Sweating a lot
- Seizures
- Numbness or tingling in the mouth
- Metallic taste
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Spinal:

- Trouble passing urine
- Loss of bladder or bowel control
- Not able to get or keep an erection
- Long-lasting burning, numbness, tingling, or paralysis in the lower half of the body
- Backache
- Fever or chills
- Stiff neck
- If bright lights bother your eyes

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Bupigobbi;
- (CH) Switzerland: Bupivacain Sintetica;
- (DO) Dominican Republic: Kolac;
- (EE) Estonia: Marcaine;
- (FI) Finland: Marcain;
- (GB) United Kingdom: Exparel liposomal;
- (ID) Indonesia: Buvanest;
- (IN) India: Bupitroy;
- (KW) Kuwait: Bucaine;
- (LB) Lebanon: Betolina;
- (LT) Lithuania: Bucain Hyperbar;
- (NL) Netherlands: Bupivacain hcl | Bupivacain hcl pch | Bupivacaine Accord | Bupivacaine hcl grindeks | Bupivacaine hcl svizera;
- (NO) Norway: Marcain | Marcaine;
- (NZ) New Zealand: Marcain;
- (PH) Philippines: Marcaine | Sensorcaine;
- (PR) Puerto Rico: Bupivacaine HCL | Sensorcaine | Sensorcaine mpf | Xaracoll;
- (QA) Qatar: Betolina | Bucaine | Bupicaine Rachianesthesia | Bupican | Bupican Heavy | Buvasin Heavy | Marcain | Marcain Spinal Heavy | Marcaine PF;
- (TW) Taiwan: Marcaine;
- (UA) Ukraine: Novostezin Spinal Heavy;
- (UG) Uganda: Anawin

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